CodeBlooded

PackagePro

Member Name	Role
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Navya Agrawal	Design
Priyanka Rajput	Frontend & UI
Priyanshi Singh	Pitch Lead

The Problem

Every traveller is different. So why does travel still come in fixed packages?

- Packages are built around the itinerary, not the traveller.
- Personalising a trip often means juggling multiple platforms, prices, and options.
- Pricing changes with customisation, but the logic behind it is rarely clear.
- Finding the right guide based on language, interests, and availability is still fragmented.
- Travellers spend more time comparing options than actually planning their journey.

The Opportunity

One intelligent platform that puts the traveller at the centre, adapting the journey, pricing, and experiences to their needs.

What we have built

- ***Discover & Personalize***

  Curated Journeys • AI Trip Builder • Personalized Discovery • History-Based Planning

- ***Customize & Compare***

  Dynamic Customization • Package Swaps • Real-Time Repricing • Smart Price Comparison • Transparent Pricing

- ***Smart Trip Planning***

  Smart Add-ons • Guide Match • Travel Pace Personalization • Budget Optimizer • Itinerary Conflict Alerts

- ***Connected Travel Experience***

  Language-Aware Travel • Interactive Trip Map • Save, Share & Book

Out of Scope

- ***Group Preferences***

  Plan trips around preferences of multiple travellers.

- ***Alternative Destinations***

  Discover other destinations matching your dates and budget.

- ***AI Guide Preview***

  Preview a guide's expertise and tour experience with AI.

Traveler's Journey

1

Set Preferences

User sets their preferred language, budget, interests, and desired travel pace

2

Discover Your Trip

Browse packages by theme
View packages personalized from preferences + booking history

3

Build with AI

AI creates a complete package using the user's preferences, history, and available travel components

4

Choose / Swap Your Package

Compare package prices
Swap between package options
Choose the package that fits best

5

Customize Your Trip

User can add, remove, or replace individual components

6

Personalize Your Experience

Select a local guide based on language & specialization
Get smart add-on recommendations
Add recommended experiences with one click

7

Optimize Your Trip

PackagePro checks the customized trip for: Budget, Travel pace,Activity conflicts
AI/system suggests changes when necessary

8

See Everything Update Live

Every change instantly updates:
Live total price
Price breakdown
Day-wise itinerary
Interactive trip map

9

Review & Share

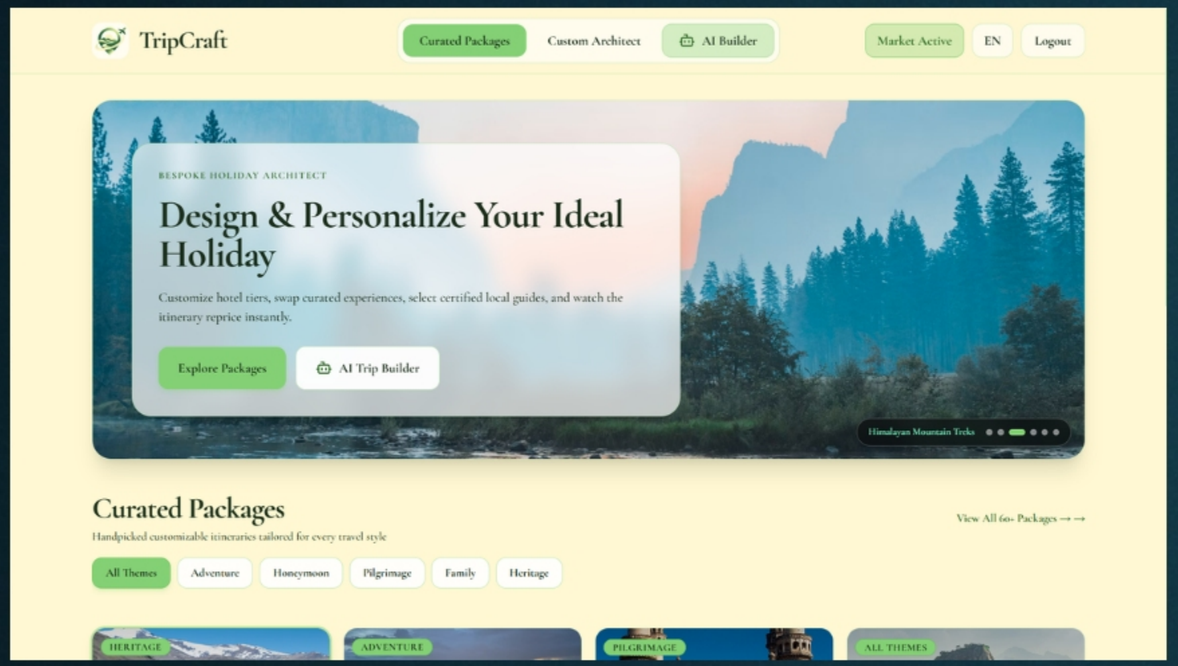
Review all your selections and then Save or Share the package

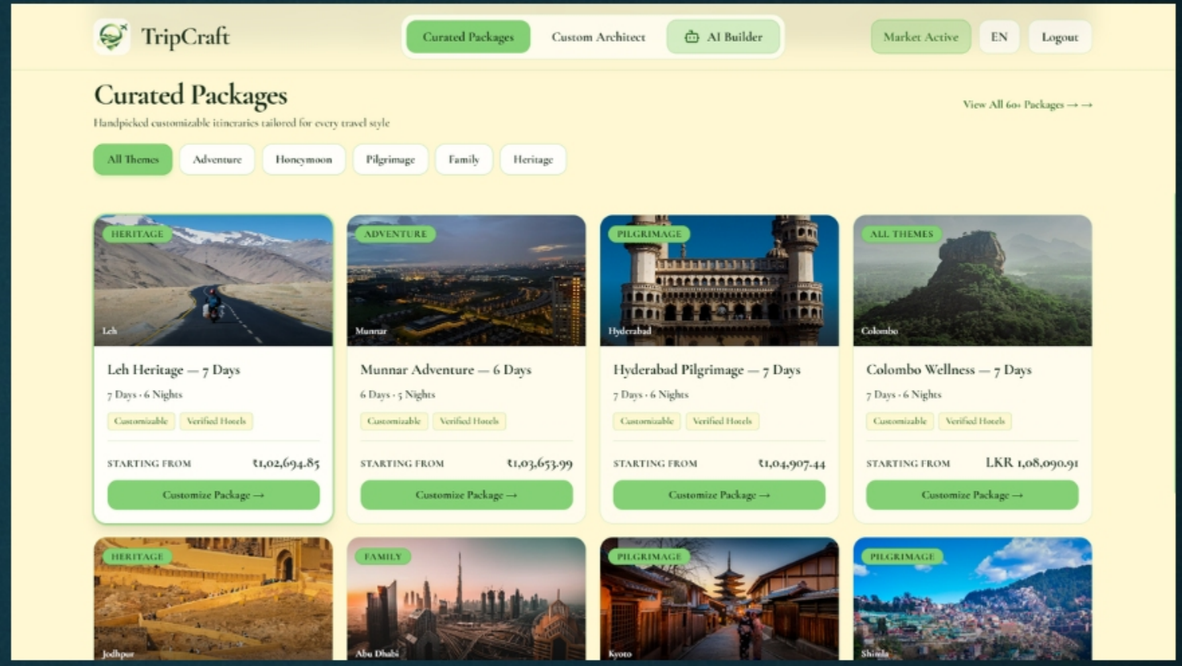
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Book Your Trip

User confirms the customized package and receives a booking confirmation.

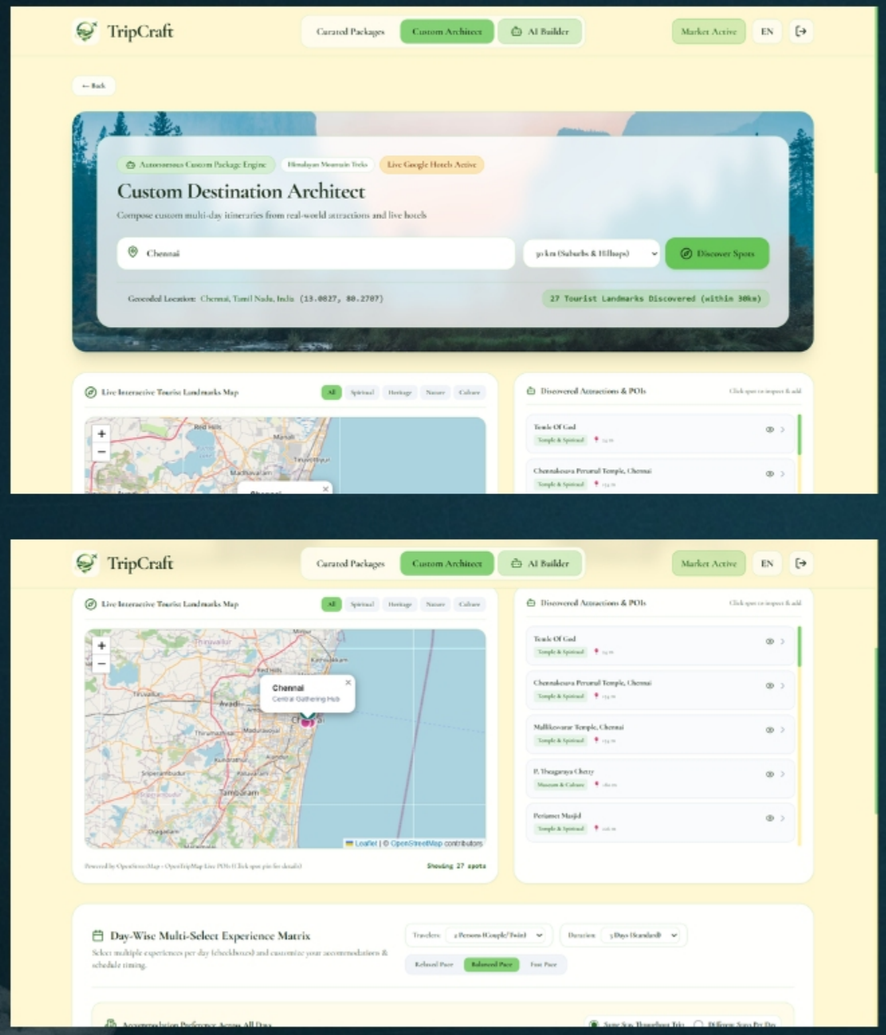
Traveler’s Screen

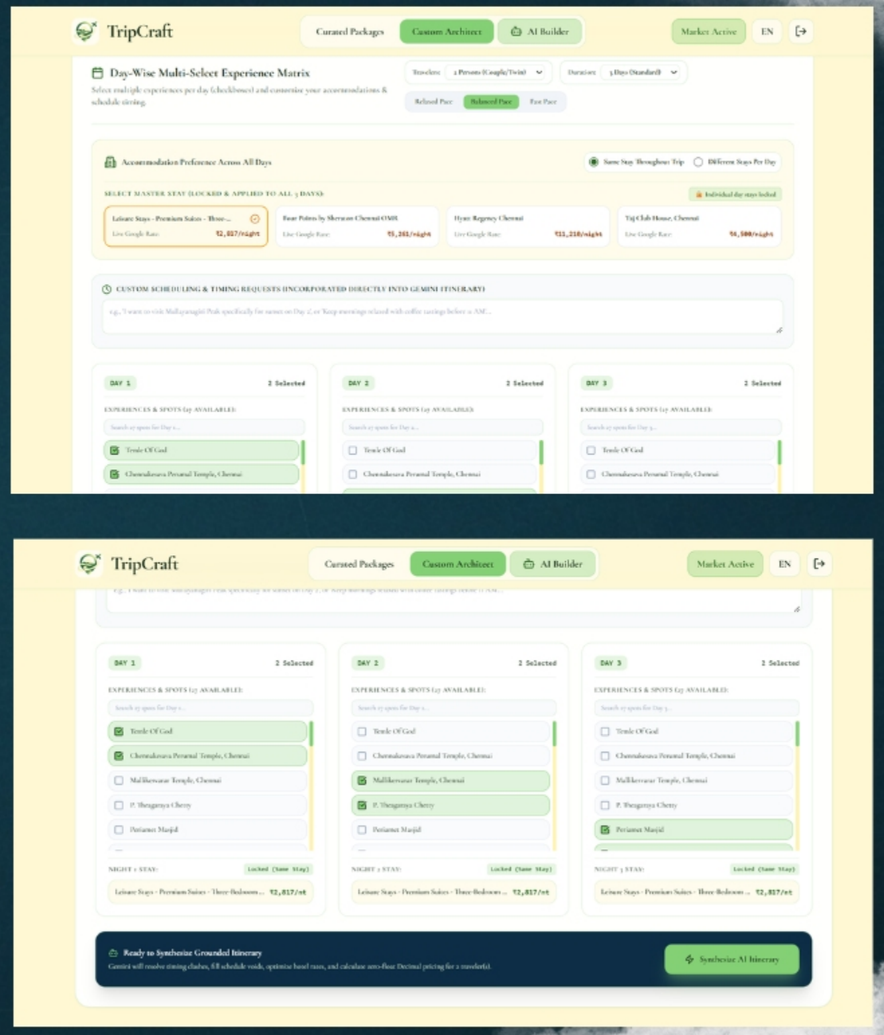




Discover Packages

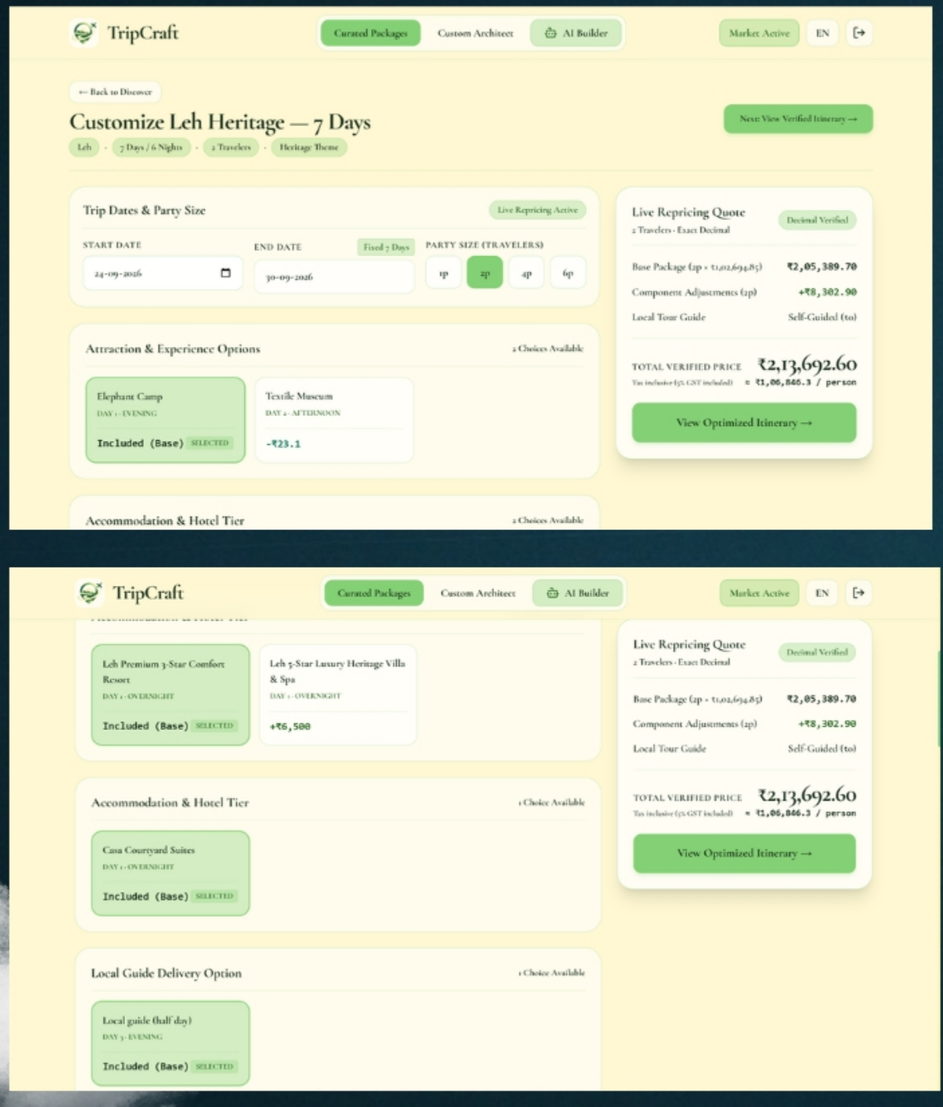
Traveler's Screen

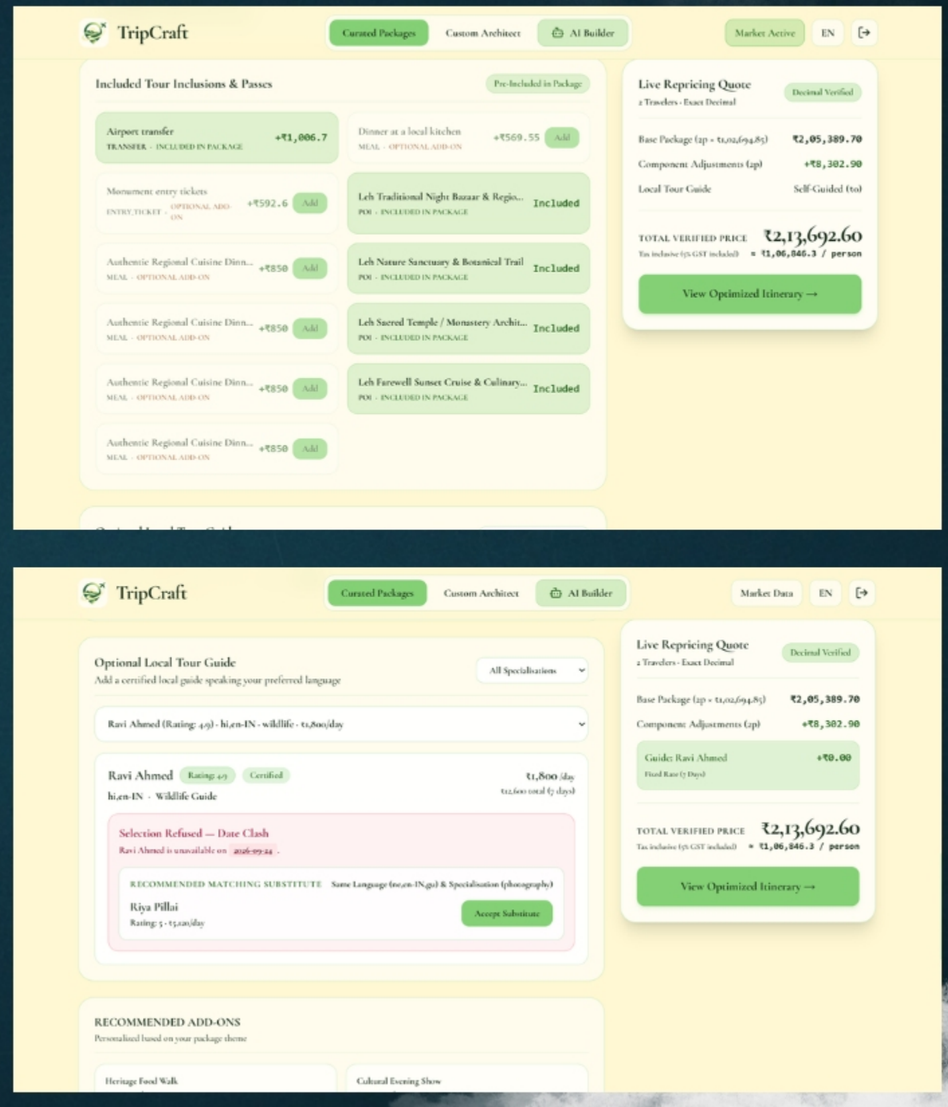




Build your package

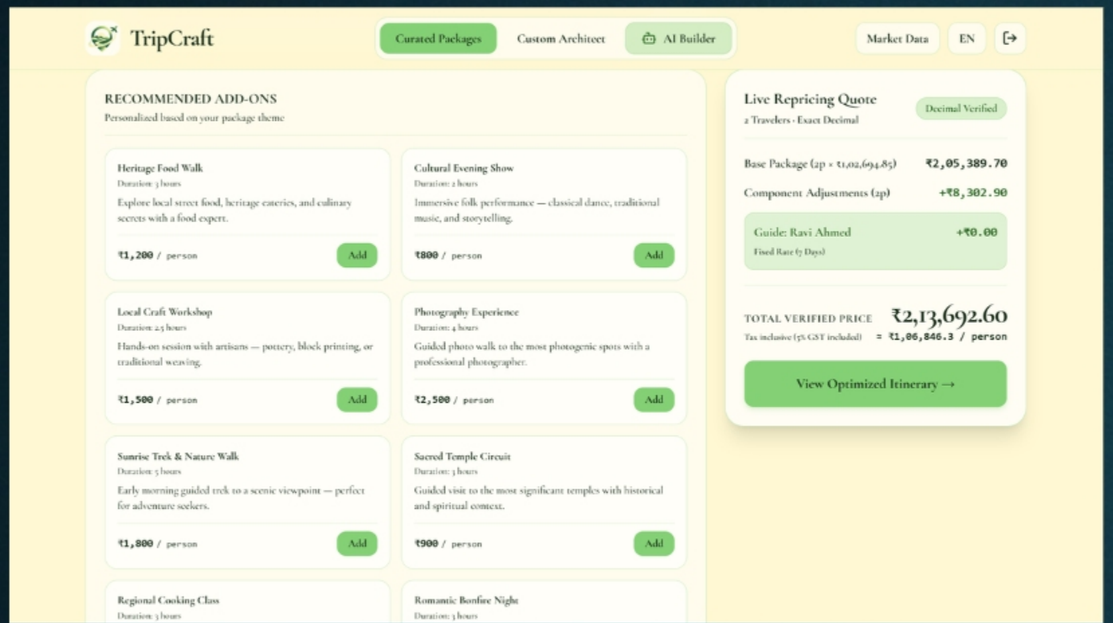
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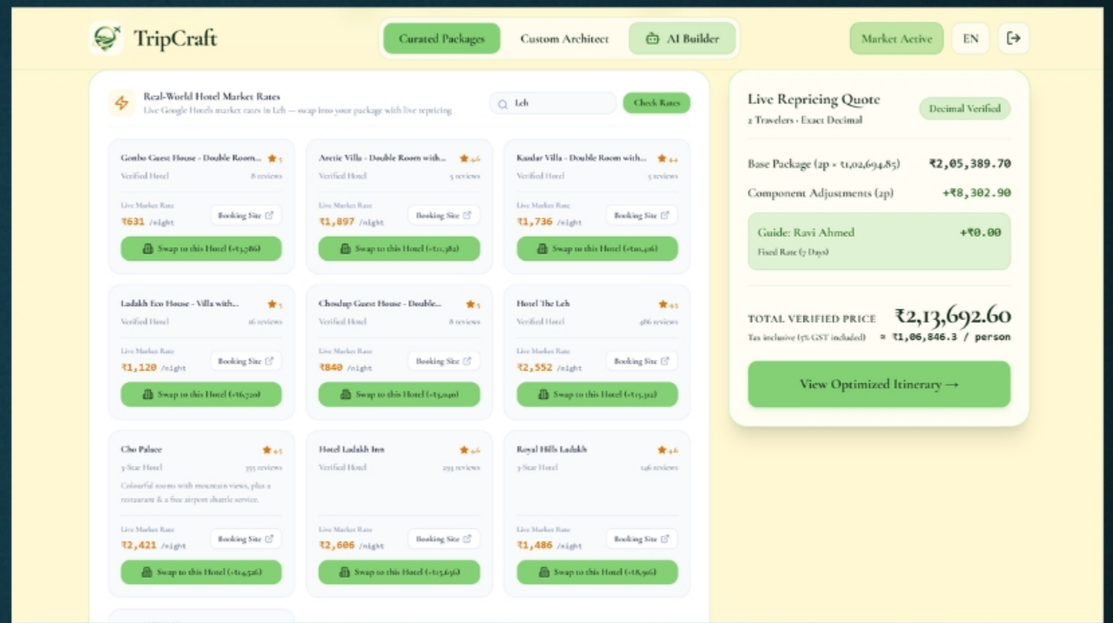




Customize packages

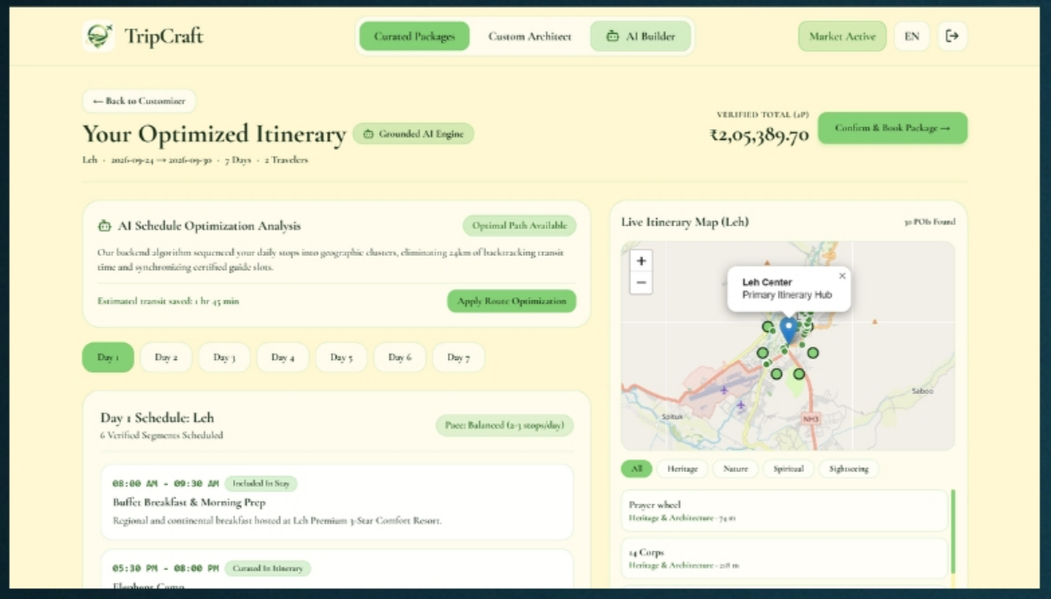
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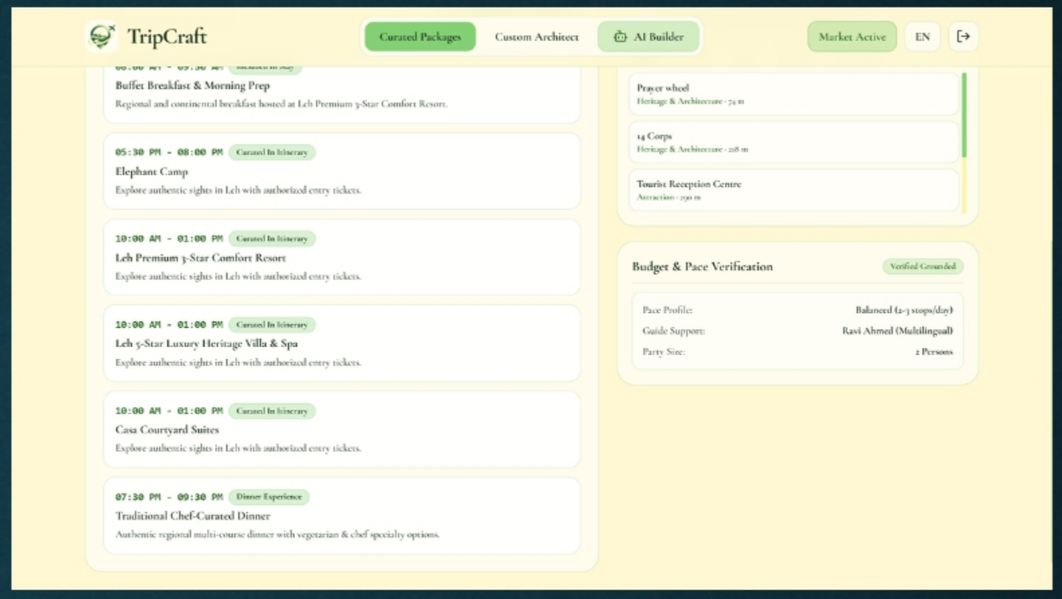




Customize packages

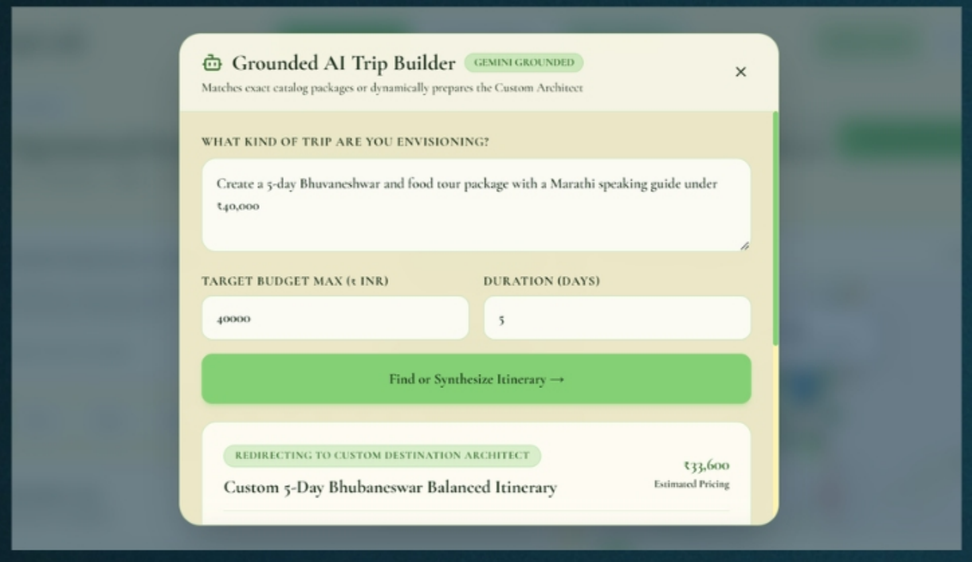
Traveler's Screen





Package optimization

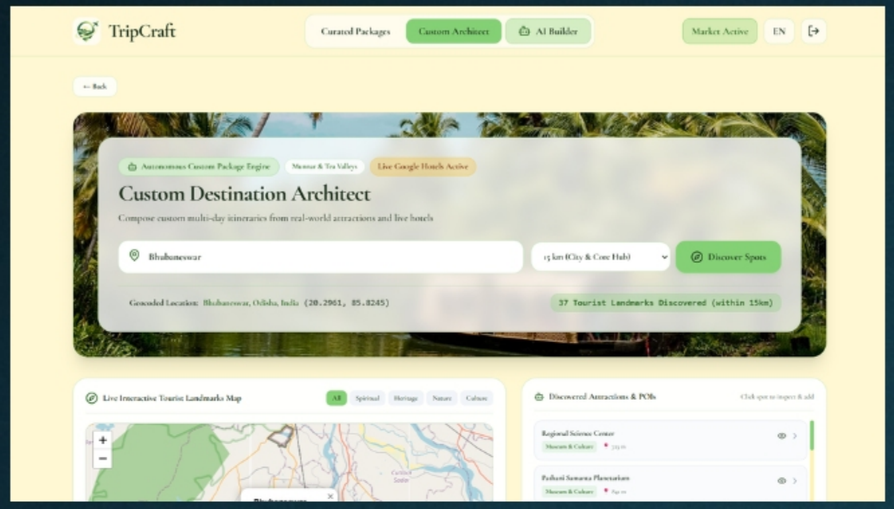
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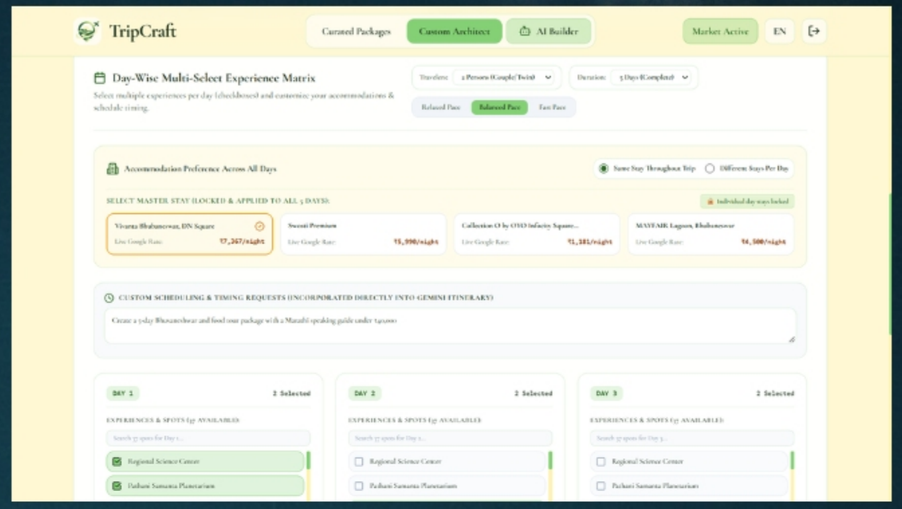




AI Builder

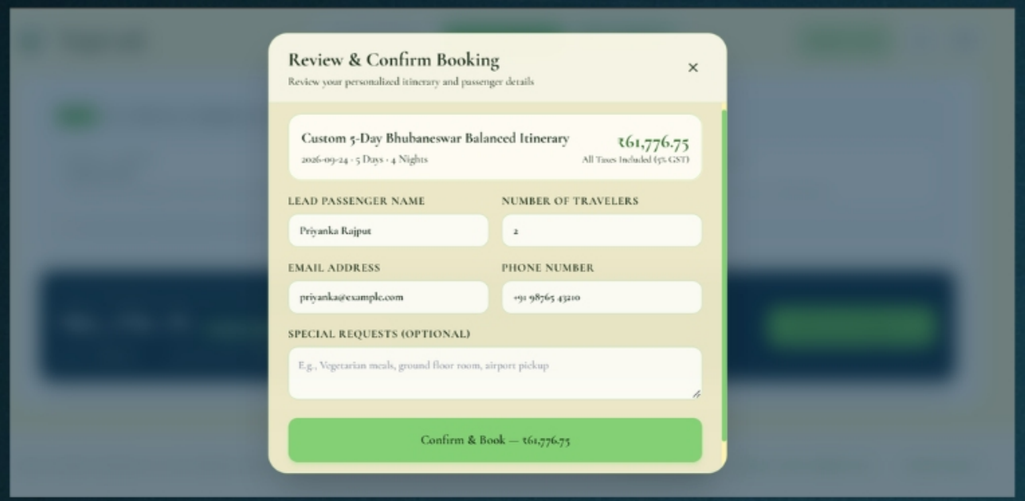
Traveler's Screen

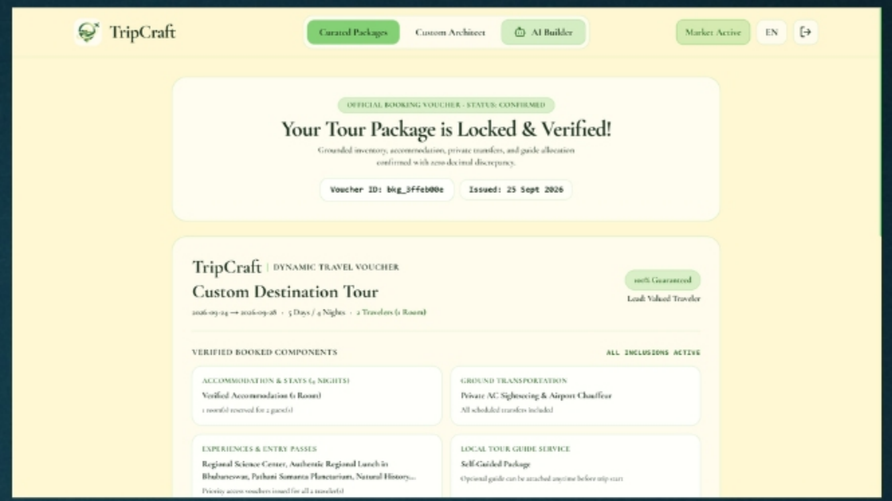




AI Builder

Traveler's Screen





Confirmation and Booking

TRIPCRAFT · AI TRAVEL PLATFORM

TripCraft System Architecture

Dynamic Tour Package Assembly · Grounded Gemini AI · Zero-Float Decimal Pricing · Mandatory Guide Availability & Clash Proofs

60+ Packages | 100% Decimal Math | Guide Clash Proof | 8 Languages (BCP-47) | 90km Geo Radius

STEP 01 · TRAVELER INTENT

Holiday Preferences

Language & Budget
- Selects UI language (en, hi, ta, etc.)
- Sets vacation budget cap
- Chooses trip duration (days)
- Sets pace (Relaxed vs Fast)

Customization Actions
- Explores 60+ pre-built packages
- Prompts Grounded AI builder
- Swaps hotel tiers & rooms
- Adds curated local experiences

Guide & Verification
- Selects local speaking guide
- Instant date clash validation
- 1-Click substitute fallback
- Live real-time quote review

User Action →

STEP 02 · FRONTEND STUDIO

TripCraft UI (React 18 + Vite)

Landing & Package Catalog
- 10s Animated showcase carousel
- 60+ packages with theme filters & opaque tags

Customizer Studio & Live Repricing
- Swappable hotel tier cards with signed deltas
- Zero-float live client state synchronization

Guide Selector & Clash Banner (§2.2)
- Red clash warning on unavailable dates
- 1-Click substitute card with live price delta

Destination Architect & Maps
- Custom city trip builder with 90km POI search
- Leaflet map marker canvas & printable PDF voucher

REST JSON →

STEP 03 · BACKEND ORCHESTRATION

TripCraft Core Engine (FastAPI)

API Gateway & Router Modules
- /api/catalog, /pricing, /guides, /ai, /bookings
- Pydantic data validation & decimal serialization

Grounded Gemini AI Pipeline
- Prompt -> Structured DB Entities (zero hallucinations)
- Strict ranking over real SQLite catalog rows

Decimal Repricing Engine (R3, §9)
- base_price + Σ(deltas) + (day_rate * days * multiplier)
- Recomputed from 0 on every call with 0.00 precision

Guide Availability & Substitute Logic
- Calendar query over (guide_id, date) range
- Same-language + same-specialisation fallback match

Live Feeds →

STEP 04 · LIVE APIS

Real-World Travel Feeds

Google Hotels (SerpAPI)
- Live real-world hotel rates
- Room options & verified pricing

OpenTripMap Attractions
- 90km Radius POI discovery
- Heritage, nature & cultural spots

Azure AI Translator v3.0
- Multilingual batch strings
- 8 Indian regional languages

Leaflet & OpenStreetMap
- Interactive spatial map pins
- Day-wise route visualization

Audits →

STEP 05 · PS-04 RULES

Governance

Rules R1–R8
- Opaque string IDs
- ISO-8601 & BCP-47

Rate Multipliers
- Peak uplifts (1.25x)
- Slot-level counts

Safe Bookings
- Idempotency keys
- Soft-delete lifecycle

17/17 Test Proofs
- Clash test passed
- Decimal math test

CANONICAL RELATIONAL DATA STORE (SQLite PS-04.db v1.1.0-rc1 · 21 Tables · 28,103 Rows)

User & Preferences Store
- users (Accounts, profiles, contact info)
- user_preferences (UI language, guide_language)
- Strict BCP-47 tag conformance (R6)

Travel Catalog & Guide Store
- tour_packages (60+ curated packages, base_price)
- package_components (hotel_tier, activity, deltas)
- tour_guides & guide_availability (calendar slots)

Trips & Versioned Itineraries
- trips (start_date, end_date, traveler_count)
- itineraries (version, is_active flag)
- itinerary_items (day slots, coordinates, places)

Selections & Booking Records
- package_selections (Additive table per R1-R8)
- bookings (Idempotent reservations, status)
- hotels, room_types, transfers, media

State synchronization & constraint verification

EXPERIENCE & PLATFORM GUARDRAIL LAYER

Realtime UI Repricing
- Fast API round-trips on every component swap
- Live Decimal quote & delta breakdown

Trust & Availability Guardrails
- Zero float drift · Hard refusal on clashing dates
- Strict BCP-47 server-side language validation

Pace & Timing Constraints
- Day-wise stop limits (Relaxed vs Packed)
- Sunset timing checks (e.g. waterfalls before dark)

Authoritative AI Guardrail
- Gemini suggests and explains;
- Database & Decimal retain 100% price authority

Confirmed holiday package outcome

TERMINAL OUTCOME (PS-04 REQUIREMENT MET)

A Fully Verified, Live-Repriced & Bookable Tour Package

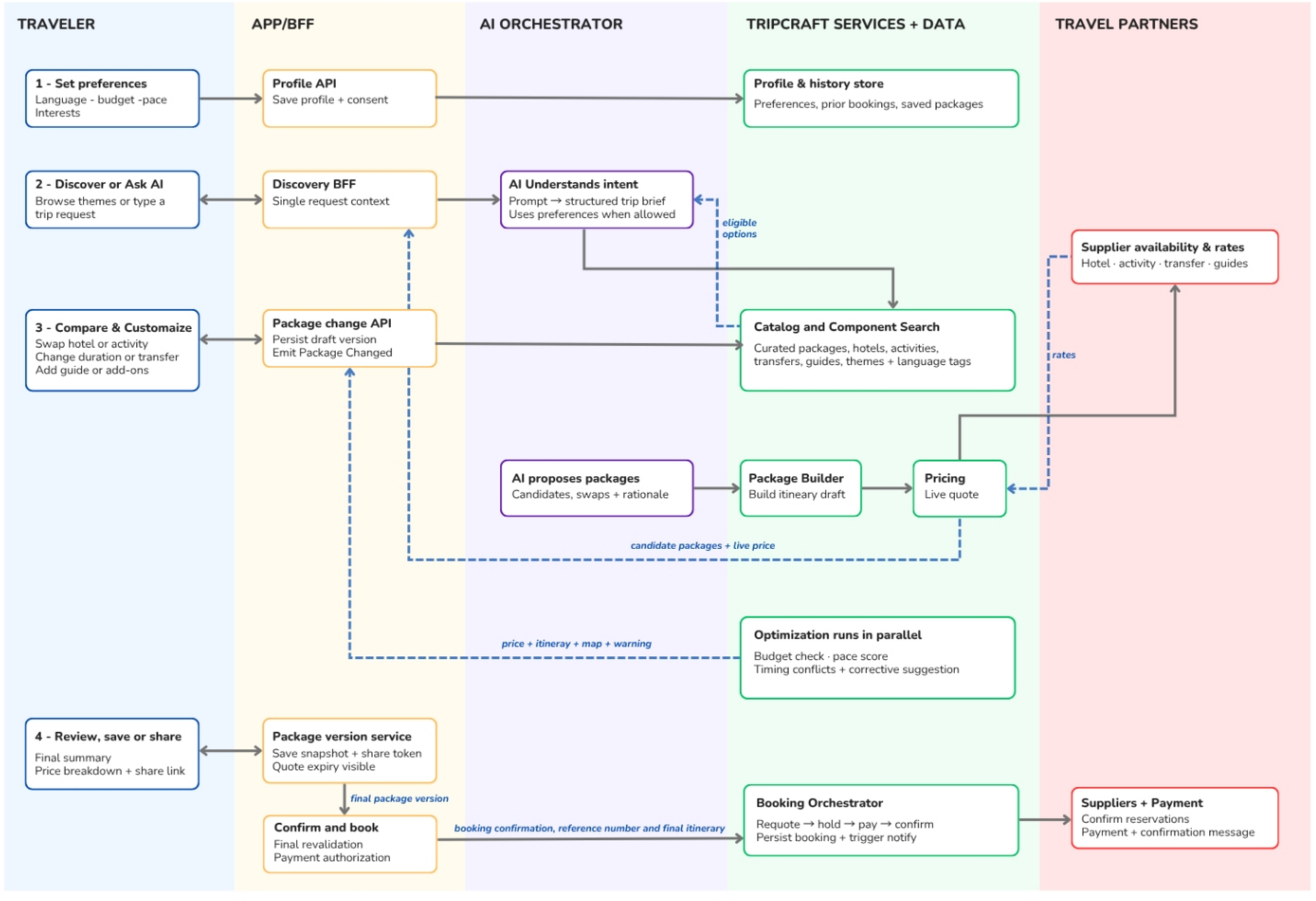
Traveller customizes a tailored holiday with exact Decimal pricing, verified calendar guides, interactive Leaflet coordinate maps, and instant official voucher receipts in their preferred regional language.

Architecture Diagram

Flow Diagram

TripCraft — user & data flow

Solid arrows are requests and decisions. Blue dashed arrows are live data returned to the traveler.



AI Features

AI TRIP BUILDER

Generate personalized packages
Grounded in: Package + component data
Measure: Preference & budget match

PERSONALIZED RECOMMENDATIONS

Use preferences + booking history
Grounded in: User profile & history
Measure: Recommendation relevance

SMART ADD-ONS

Recommend relevant experiences
Grounded in: Interests + package data
Measure: Add-on acceptance

GUIDE MATCHING

Rank the best-fit local guides
Grounded in: Language + specialty + price
Measure: Match accuracy

BUDGET OPTIMIZATION

Suggest alternatives to stay on budget
Grounded in: Component pricing + budget
Measure: Budget adherence

Business Value

From Browsing to Booking

Personalized trips reduce decision fatigue and turn more travellers into confirmed bookings.

Every Inventory, Better Utilized

Alternative hotels, activities and guides help businesses utilise available inventory.

One Trip, More Value

Smart add-ons, upgrades, guides and experiences increase the value of every booking.

Transparent Pricing, Greater Trust

Live repricing and component-level costs eliminate surprises and build confidence.

Personalization That Compounds

Every preference and booking makes future recommendations more relevant.

Personalization at Scale

AI automates package creation and modification, allowing businesses to scale without extra effort.

Tech Stack

Backend	FastAPI	Accurate price calculations with Decimal, plus fast API development and automatic documentation.
Frontend	React + Vite	Makes the interface interactive and provides fast development with instant reloads.
Database	SQLite	Uses the provided PS-04.db directly, so no time is wasted on database migration.
Styling/UI	Tailwind CSS + shadcn/ui	Helps us create a clean and consistent UI quickly.
LLM/AI	Gemini API	Ranks suitable packages from real database results instead of generating packages from scratch.
Maps	React-Leaflet + OpenStreetMap	Free interactive maps using the existing location data.
APIs	SerpAPI, OpenTripMap, Azure Translations	APIs for hotels, tourist attractions and for multilingual translation

Multilingual by Design

Where it appears

Surface	Data Used	Effect
Interface	users.locale	Changes basic UI text
Guide selection	user_preferences.guide_language	Shows guides who speak the selected language
Package recommendations	tour_packages.languages_offered	Filters/ranks packages based on language

How we handle it

Storage	Languages are stored using standard BCP-47 language codes and validated through the languages table.
Filtering	Language preference is applied on the server to filter both guides and packages – it is a functional preference, not just a UI translation.
Fallback	If no match is available, we show the English option with a clear note instead of showing an empty result.